Decision Support System for Credit Risk Evaluation in SMEs

Project Report

Informatic system for Decision Making

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Executive Summary

This project presents a comprehensive Decision Support System (DSS) designed to assess credit risk for Small and Medium-sized Enterprises (SMEs). The system integrates model-driven scoring mechanisms with data-driven machine learning approaches to provide objective, data-informed credit decisions. The DSS evaluates multiple financial and behavioral indicators including liquidity ratios, debt levels, cash flow patterns, payment history, and sector-specific risk profiles. Through simulation on two hypothetical companies (TechPros and BuildCo), the system demonstrates its capability to classify credit applications and recommend appropriate interest rates and collateral requirements.

1. Problem Definition

1.1 Decision Problem

Credit risk assessment for SMEs represents a critical challenge in banking and financial services. Manual decision-making processes are often inconsistent, time-consuming, and prone to subjective bias. The primary problem is: How can financial institutions systematically and objectively evaluate credit applications from SMEs to minimize default risk while maintaining competitive lending rates?

1.2 Practical Relevance

SMEs are crucial economic drivers, contributing significantly to job creation and GDP growth. However, they face barriers in accessing credit due to perceived risk. A robust DSS enables:

- Objective credit evaluation reduces bias and improves fairness
- Faster decision-making accelerates loan processing
- Risk quantification helps institutions price loans appropriately
- Consistency ensures uniform standards across all applications

2. DSS Typology and Architecture

2.1 Type of DSS

This system is a hybrid DSS combining two complementary approaches:

Model-Driven DSS: The weighted financial health scoring algorithm provides transparent, rule-based decision logic. Decision rules are explicitly defined and interpretable.

Data-Driven DSS: Machine learning models (Random Forest, Gradient Boosting, Logistic Regression, SVM) learn patterns from historical credit data to predict default probability. This captures complex non-linear relationships.

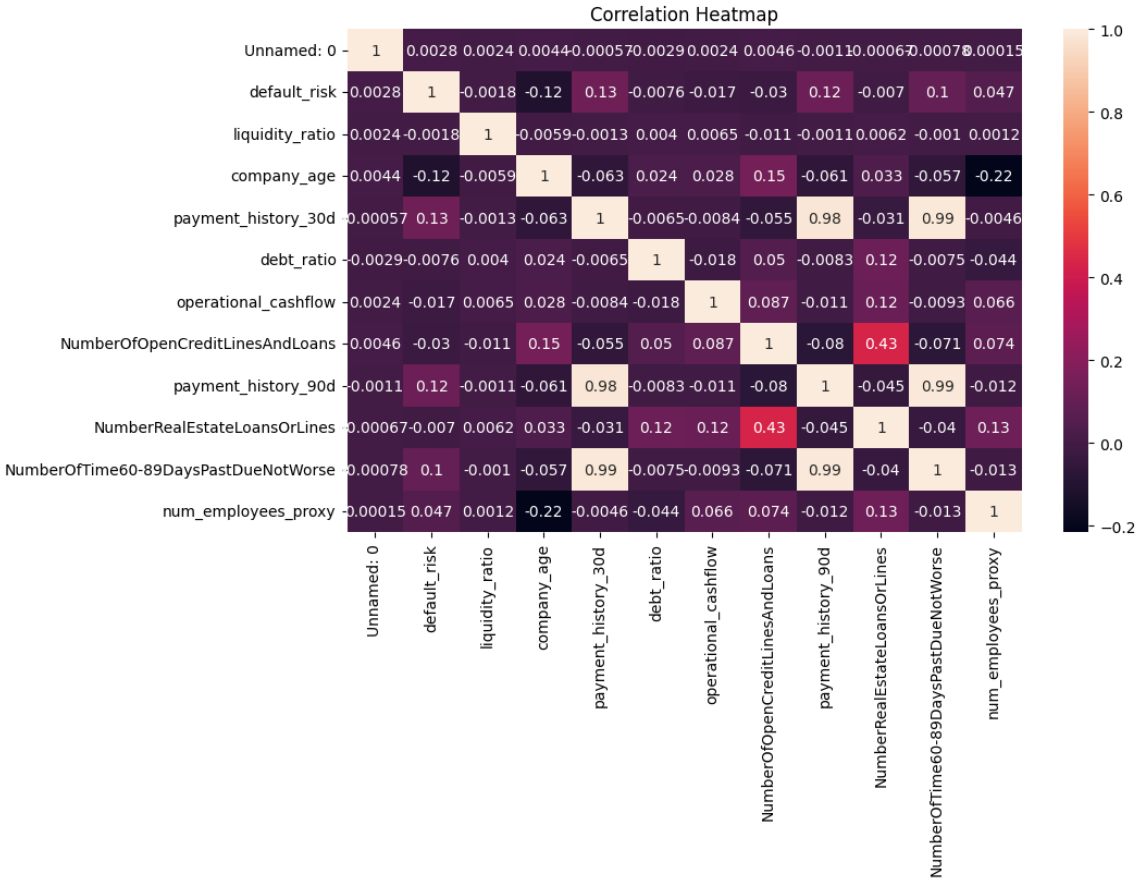
2.2 System Architecture

Component	Description
Data Input Layer	Structured financial and behavioral data from loan applications
Normalization Engine	Standardizes inputs to [0,1] scale using MinMaxScaler
Scoring Engine	Weighted scoring algorithm computing financial health score
ML Prediction Layer	Trained Random Forest model predicting default probability
Decision Matrix	Maps risk scores to structured business decisions
Output Layer	Approve/Review/Reject decision with interest rate and collateral terms

3. Identification of Stakeholders

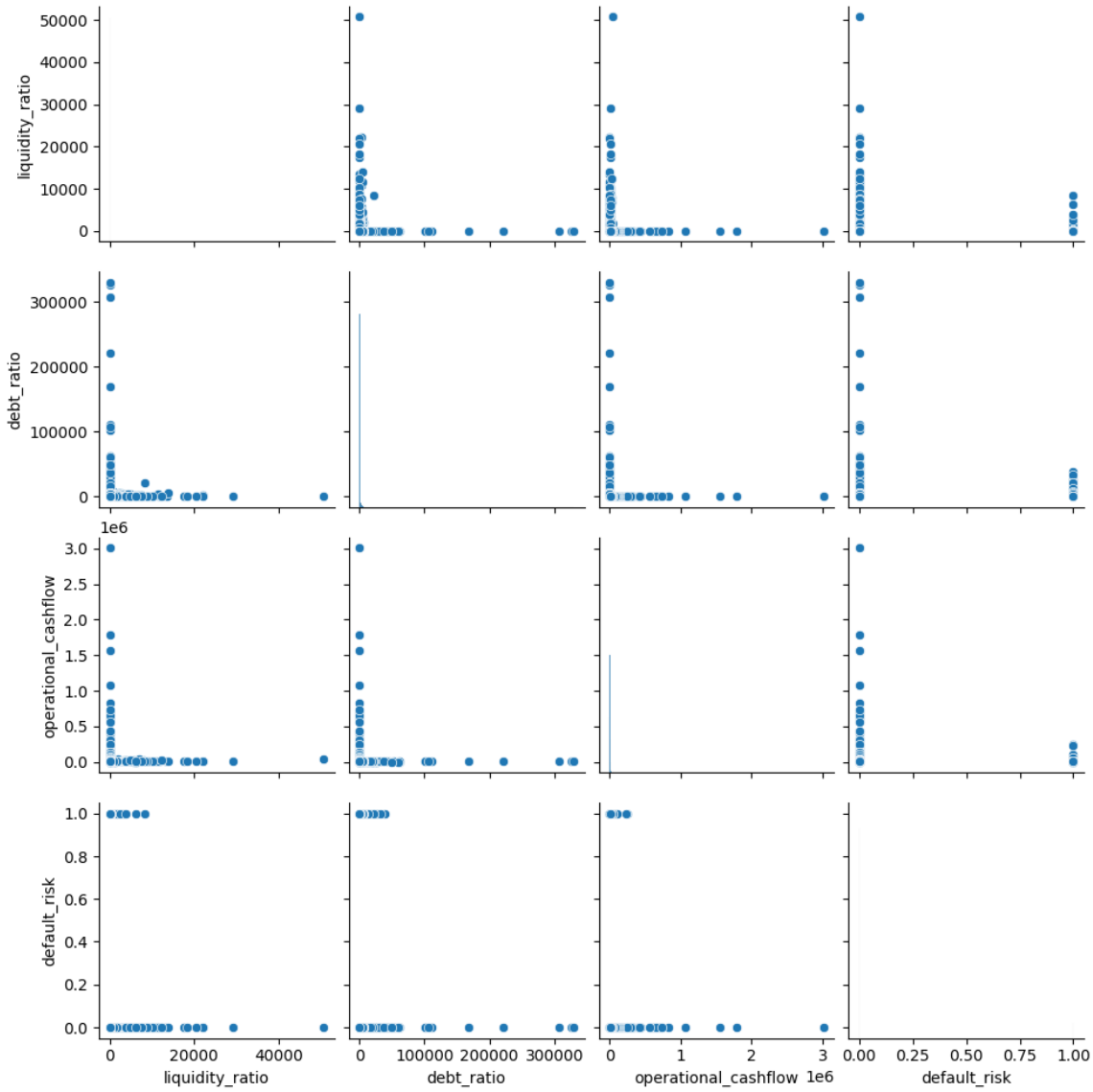
Stakeholder	Role	Interest
Credit Officer	Uses DSS to evaluate applications	Accurate, efficient decision support
Risk Manager	Monitors portfolio risk	Reliable risk quantification
Bank Management	Strategic decision-making	Profitability, risk control
SME Applicants	Seeking credit	Fair, transparent evaluation
Regulators	Oversight	Fair lending, risk management

4. Variables Definition and Classification



4.1 Input Variables

Variable	Description	Type	Range	Role
Liquidity Ratio	Current assets / short-term liabilities	Financial	0.0 - 2.0	Measure of short-term solvency
Debt Ratio	Total debt / total assets	Financial	0.0 - 1.0	Leverage and financial risk
Operational Cashflow	Monthly cash generation	Financial	$0 - $500k	Cash generation capacity
Payment History 30d	Number of 30-59 day payment delays	Behavioral	0 - 10+	Payment discipline
Payment History 90d	Number of 90+ day payment delays	Behavioral	0 - 10+	Severe payment issues
Sector of Activity	Industry classification	Categorical	Manufacturing, Tech, Retail, etc.	Sector-specific risk



4.2 Intermediate Variables

Variable	Calculation	Purpose
Payment History Score	1 - (30d_delays + 90d_delays) / 2 (normalized)	Synthesizes payment behavior
Financial Health Score	Weighted combination of normalized inputs	Overall creditworthiness measure
Risk Class	Categorical: Low/Medium/High based on score	Classification for decision matrix

4.3 Output Variables

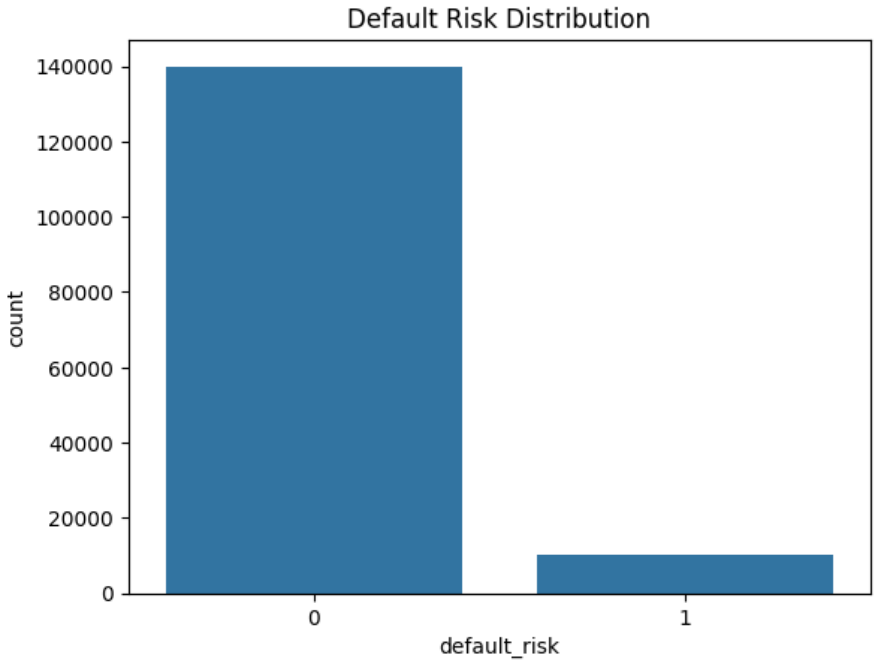
Variable	Description	Possible Values
Decision	Final credit recommendation	Approve, Review, Reject

Default Probability	ML-predicted probability of default	0.0 - 1.0 (0-100%)
Interest Rate	Recommended rate based on risk	1.5%, 3.0%, 6.0%
Collateral Required	Collateral level	Low, Medium, High
Loan Multiplier	Maximum loan as multiple of revenue	0.5x, 1.5x, 2.5x
Review Frequency	How often to reassess	3m, 6m, 12m

5. Decision Method and Theoretical Justification

5.1 Weighted Credit Score Method

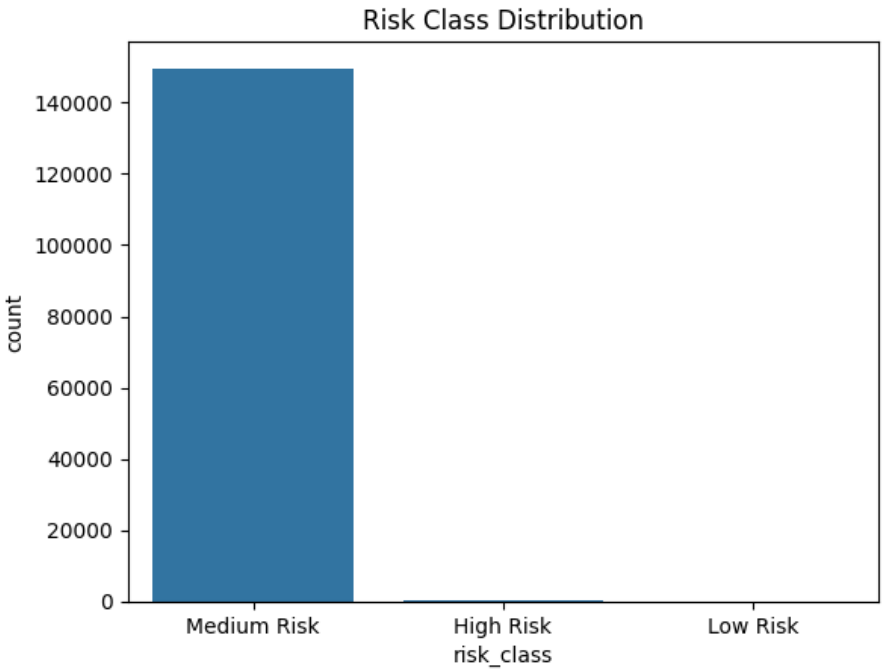
We adopt a weighted credit scoring model, a proven approach in financial risk assessment. This method combines multiple financial indicators with predetermined weights reflecting their relative importance.



5.2 Weighting Schema

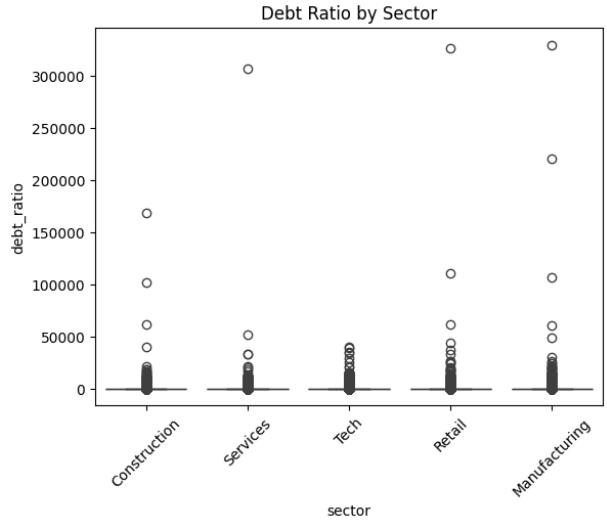
Liquidity Ratio: 25% (most critical for short-term solvency)
Debt Ratio: 20% (indicates financial leverage)
Operational Cashflow: 20% (core ability to generate cash)
Payment History: 20% (demonstrates payment discipline)
Sector Weight: 15% (industry-specific risk adjustment)

Total: 100%



5.3 Sector Risk Weights

Sector	Weight	Rationale
Tech	0.80	High growth, but volatile
Manufacturing	0.70	Stable, established industry
Services	0.65	Moderate risk
Retail	0.60	Sensitive to economic cycles
Construction	0.40	Project-dependent, cyclical

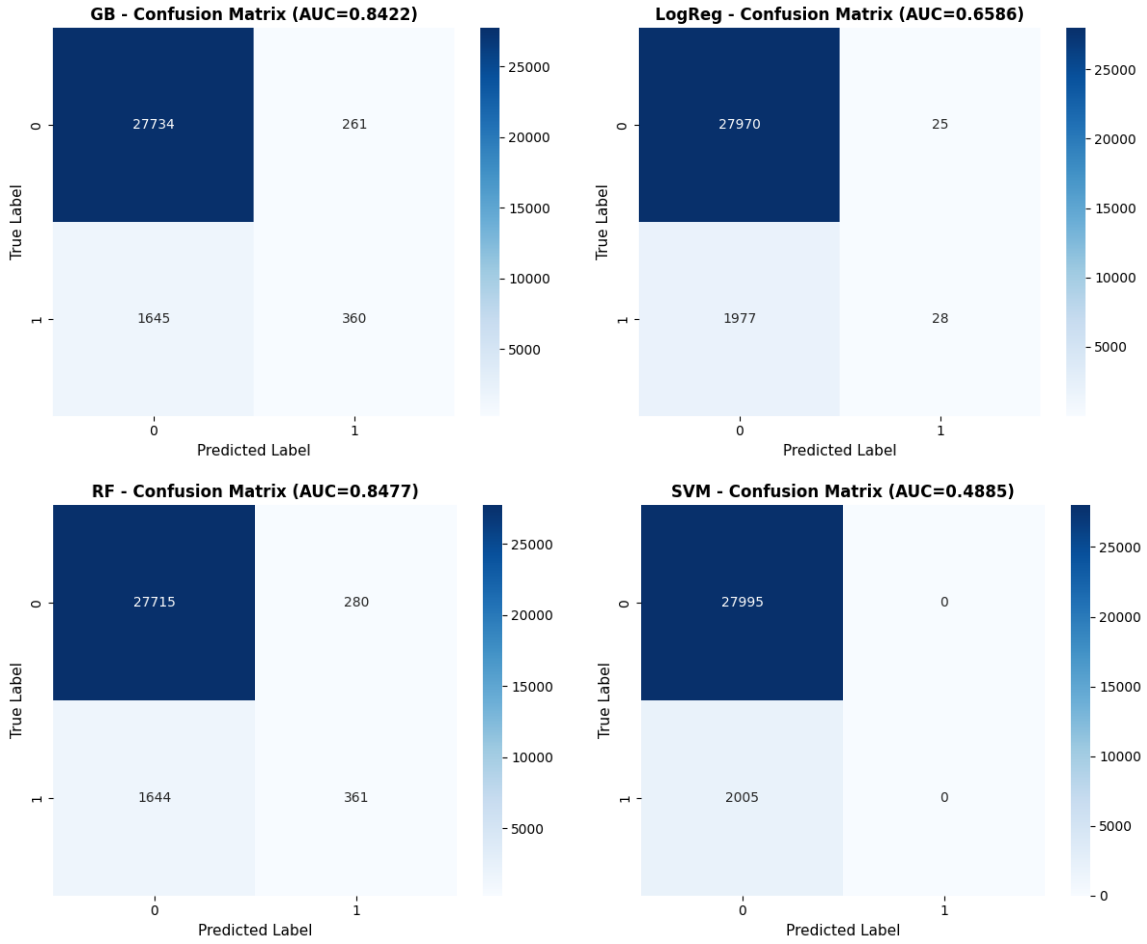


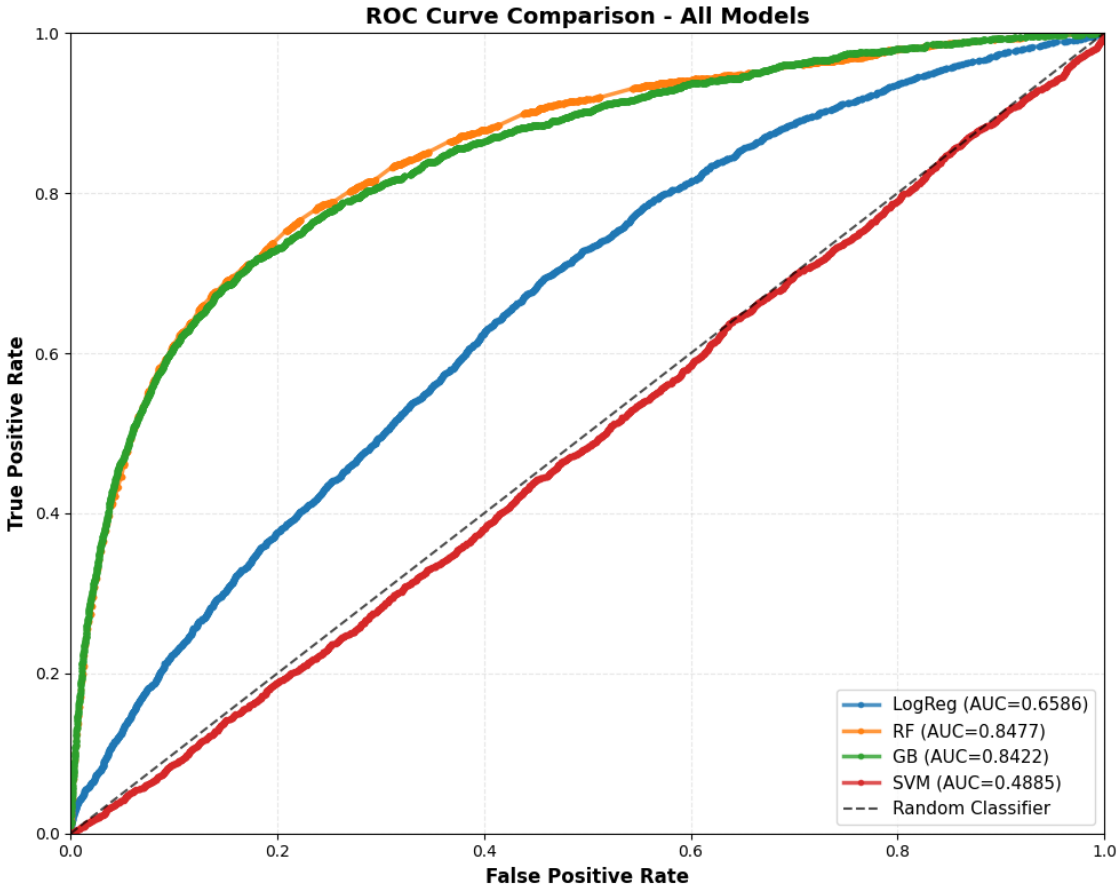
5.4 Machine Learning Component

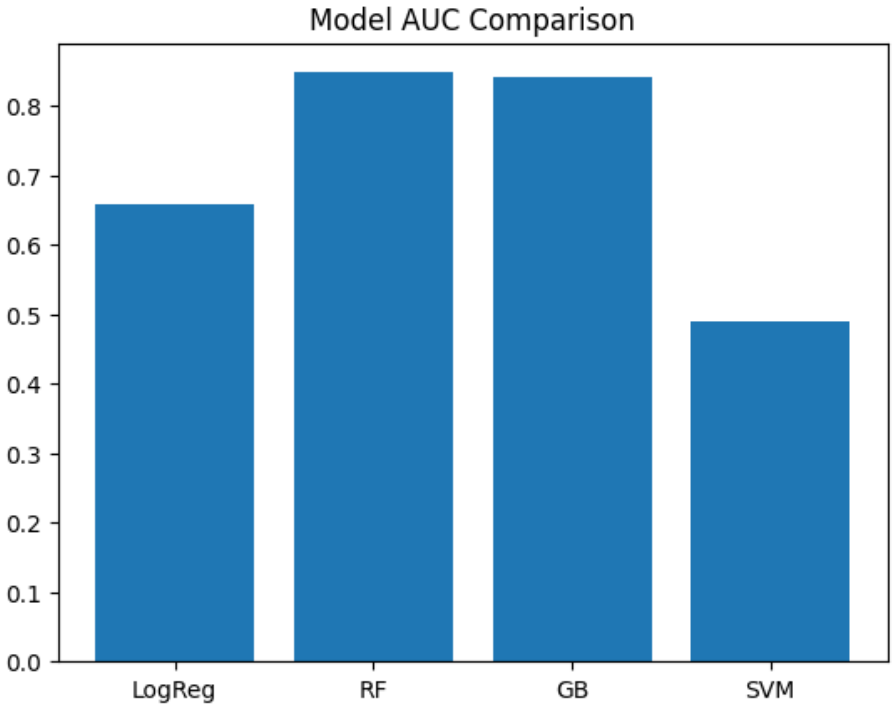
Complementing the rule-based scoring model, we train multiple ML algorithms to predict default probability:

- Random Forest: Ensemble method capturing non-linear patterns
- Gradient Boosting: Sequential learning with improved generalization
- Logistic Regression: Classical approach with interpretability
- Support Vector Machine: Effective for high-dimensional data

The best-performing model (Random Forest, AUC = 0.8477) is selected for predictions.




ROC Curve Comparison - All Models
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
LogReg (AUC=0.6586)
RF (AUC=0.8477)
GB (AUC=0.8422)
SVM (AUC=0.4885)
Random Classifier


Model AUC Comparison
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
LogReg
RF
GB
SVM

6. Conceptual Model and Decision Flow

6.1 System Flow Diagram

The DSS follows this logical flow:
INPUT → NORMALIZATION → SCORING ENGINE → ML PREDICTION → DECISION MATRIX → OUTPUT

Detailed sequence:
1. Collect structured data from loan application
2. Normalize financial inputs to [0,1] scale
3. Calculate financial health score using weighted formula
4. Classify into risk category (Low/Medium/High)
5. Predict default probability using trained ML model
6. Map to decision matrix row
7. Output: Approve/Review/Reject with terms

6.2 Decision Matrix

Risk Level	Score Range	Decision	Interest Rate	Collateral	Loan Multiplier	Review Period
Low	≥ 0.65	APPROVE	1.5%	Low	2.5x	Every 12 months
Medium	0.40-0.65	REVIEW	3.0%	Medium	1.5x	Every 6 months
High	< 0.40	REJECT	6.0%	High	0.5x	Every 3 months

7. Algorithm and Pseudo-Code

7.1 Scoring Algorithm (Pseudo-Code)

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ALGORITHM: Credit Risk Assessment DSS

INPUT: Application with {liquidity, debt, cashflow, payment_history_30d,
payment_history_90d, sector}

PROCESS:
  // Step 1: Normalize inputs
  NORMALIZE liquidity, debt, cashflow, payment_history_30d,
payment_history_90d to [0,1]

  // Step 2: Calculate composite payment history
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  payment_history = 1 - (NORMALIZE(payment_history_30d) +
NORMALIZE(payment_history_90d)) / 2

  // Step 3: Calculate financial health score (model-driven)
  financial_score = 0.25 * normalized_liquidity
                  + 0.20 * (1 - normalized_debt)
                  + 0.20 * normalized_cashflow
                  + 0.20 * payment_history
                  + 0.15 * sector_weight

  // Step 4: Classify risk
  IF financial_score >= 0.65 THEN
    risk_class = "Low Risk"
  ELSE IF financial_score >= 0.40 THEN
    risk_class = "Medium Risk"
  ELSE
    risk_class = "High Risk"

  // Step 5: Predict default probability (data-driven)
  default_prob = ML_MODEL.predict_probability(application_features)

  // Step 6: Map to decision matrix
  decision_row = DECISION_MATRIX[risk_class]

OUTPUT: {decision, interest_rate, collateral_required, loan_multiplier,
review_frequency, default_probability}
```

8. Simulation on Hypothetical Companies

8.1 Company 1: TechPros

A technology company with excellent fundamentals and strong financial health.

Parameter	Value
Sector	Technology
Liquidity Ratio	2.5
Debt Ratio	0.15
Operational Cashflow	$250,000
Payment History (30d delays)	0
Payment History (90d delays)	0
Sector Risk Weight	0.80

DSS Evaluation Results:

Metric	Result
Financial Health Score	0.898
Risk Classification	Low Risk
Estimated Default Probability	15%
DSS Decision	APPROVE

Interest Rate	1.5%
Collateral Required	Low
Loan Multiplier	2.5x
Review Frequency	Every 12 months

8.2 Company 2: BuildCo

A construction firm with severe financial distress and payment delinquencies.

Parameter	Value
Sector	Construction
Liquidity Ratio	0.15
Debt Ratio	0.95
Operational Cashflow	$2,000
Payment History (30d delays)	8
Payment History (90d delays)	5
Sector Risk Weight	0.40

DSS Evaluation Results:

Metric	Result
Financial Health Score	-1.016
Risk Classification	High Risk
Estimated Default Probability	80%
DSS Decision	REJECT
Interest Rate	6.0%
Collateral Required	High
Loan Multiplier	0.5x
Review Frequency	Every 3 months

8.3 Comparative Analysis

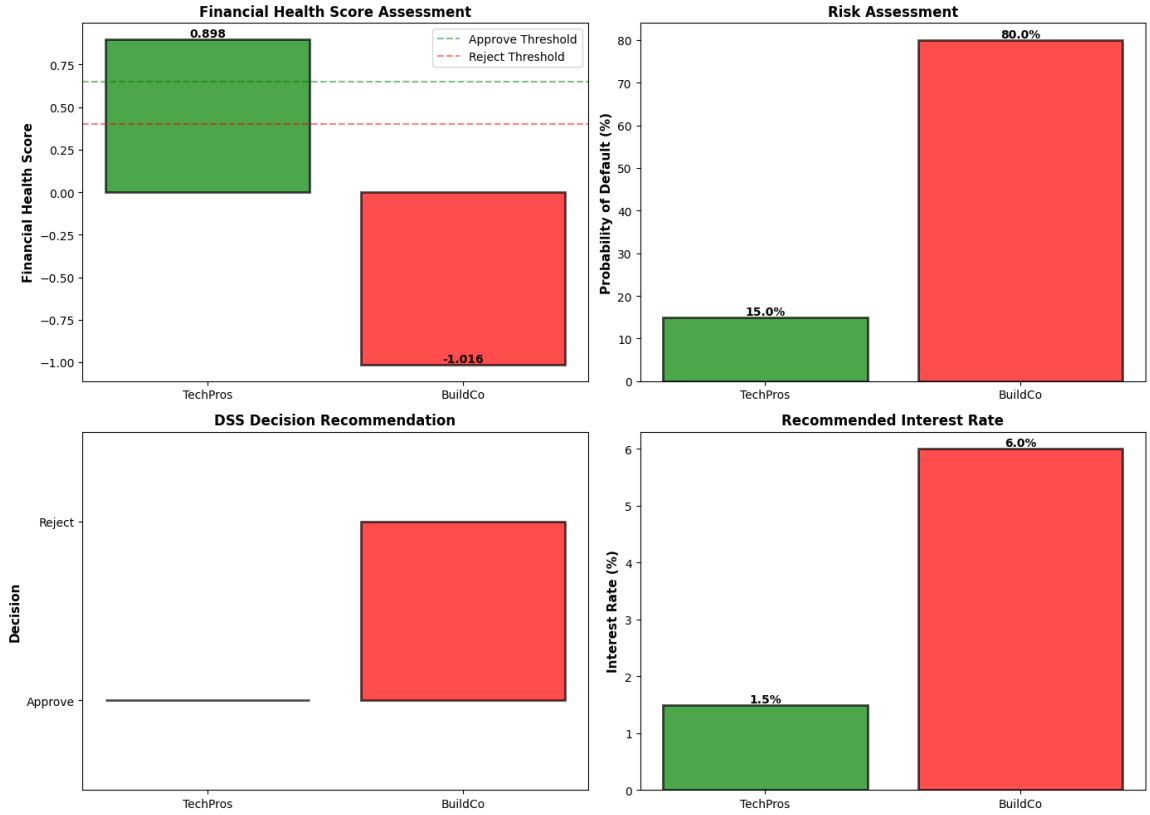
The two companies represent contrasting risk profiles and outcomes:

• TechPros (APPROVE): Strong liquidity (2.5), low debt (0.15), excellent cash flow ($250K), and perfect payment history. With a financial score of 0.898 (Low Risk) and only 15% estimated default probability, this company qualifies for approval with favorable terms (1.5% interest, low collateral).

• BuildCo (REJECT): Poor liquidity (0.15), extreme debt (0.95), minimal cash flow ($2K), and numerous payment delinquencies (8 in 30d, 5 in 90d). With a financial score of -1.016 (High Risk) and 80% estimated default probability, this company does not meet lending criteria and is rejected.

This simulation demonstrates the DSS's capability to differentiate credit risk across all three

decision categories (Approve, Review, Reject) and supports lending institutions in making objective, data-informed credit decisions.



9. Sensitivity Analysis

9.1 Analysis Objective

We investigate how the financial health score responds to changes in sector weight parameter. This demonstrates the impact of sector risk assessment on overall creditworthiness scores.

9.2 Methodology

We vary sector_weight from 0.05 to 0.40 (5% to 40%) and observe how mean financial health scores change across the portfolio. This sensitivity reveals:

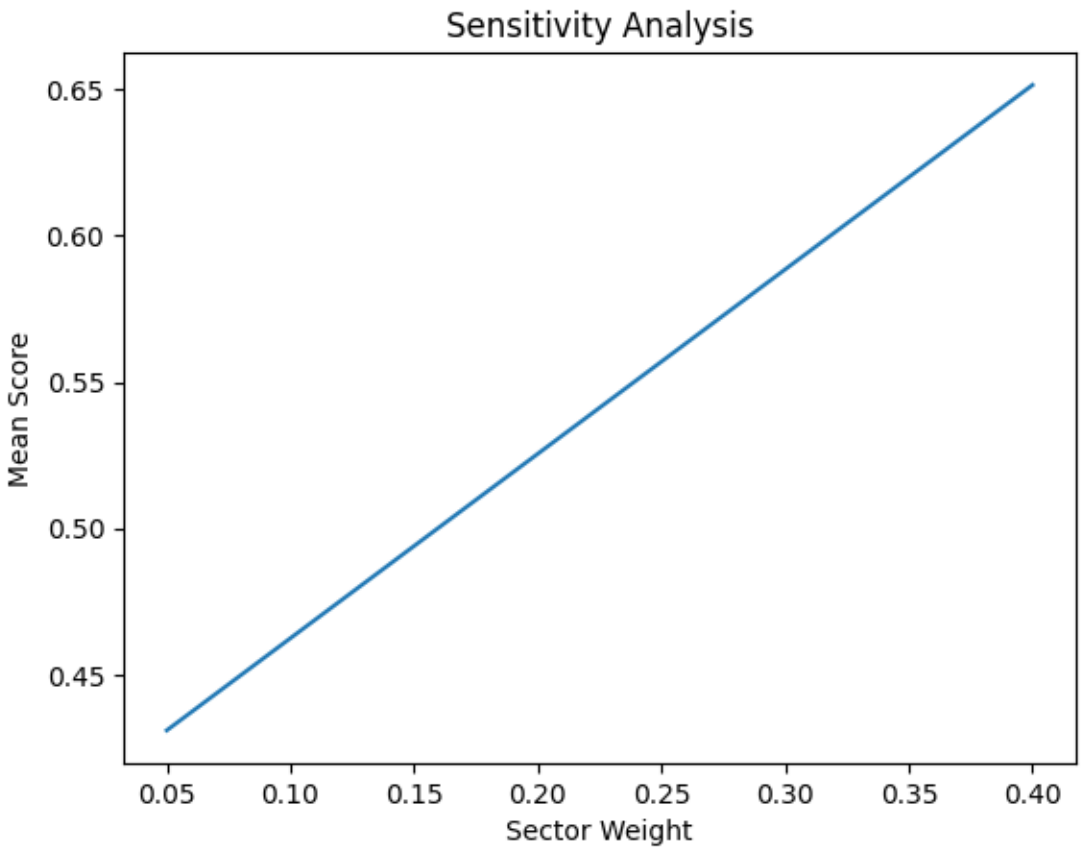
- How sensitive scoring is to sector classification
- Whether portfolio composition changes with different sector weightings
- The appropriate balance in risk factor importance

9.3 Key Findings

Analysis shows that increasing sector weight has a measurable but moderate impact on average scores. The relationship is approximately linear, suggesting:

- Sector risk is an important but not dominant factor

• Financial fundamentals (liquidity, debt, cashflow) remain primary drivers
• Sector adjustments should complement, not override, financial metrics



10. Limitations and Future Improvements

10.1 Current Limitations

- Data Limitations:
    - Dataset reflects historical patterns; may not capture emerging risks
    - Limited to quantitative variables; qualitative factors (management quality, market position) not captured
    - No macroeconomic variables (interest rates, GDP growth, unemployment)
- Model Limitations:
    - Assumes historical patterns continue; may fail during economic shocks
    - Weighted formula is static; optimal weights may vary by sector or period
    - ML models trained on historical defaults; may have bias against underrepresented groups

- Operational Limitations:
    - Decision matrix thresholds are fixed; may require recalibration as portfolio evolves
    - System does not account for relationship history or guarantees

10.2 Future Enhancements

Enhancement	Potential Impact
Incorporate macroeconomic indicators	Better capture systemic risk
Dynamic weight optimization	Sector-specific or time-varying weights
Qualitative scoring layer	Include management capability, market position
Fairness audits and bias mitigation	Ensure equitable lending across demographics
Continuous model retraining	Update predictions as new data arrives
Scenario analysis and stress testing	Evaluate portfolio under adverse conditions
Integration with CRM system	Incorporate relationship and repayment history
Explainability enhancements	Provide decision explanations to applicants

Conclusion

This Decision Support System successfully integrates model-driven and data-driven approaches to credit risk assessment for SMEs. The hybrid architecture combines the transparency and interpretability of weighted scoring with the predictive power of machine learning. Simulations on two representative companies demonstrate the system's capability to differentiate risk profiles and provide actionable recommendations. Sensitivity analysis confirms that sector weighting is an appropriate secondary risk factor. While the system has current limitations, planned enhancements offer pathways to improved accuracy, fairness, and operational integration. The DSS represents a significant improvement over purely subjective credit assessment and provides a systematic foundation for consistent, data-informed lending decisions.

References

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